Appendix A. Supplementary data

TaoChronos: A Reliable Agent for Knowledge Discovery in Traditional Chinese Medicine

TaoChronos (道乃久)

Contents: Supplementary Notes 1–4; Supplementary Tables 1–12 (each also as CSV in `tables/`); Supplementary Fig. 1; Supplementary Data 1 (the Discovery Report of the end-to-end run) and 2 (its event log).

Supplementary Note 1 | Evaluation design

TaoChronos-Eval has forty suites in three layers. The first tests the harness's boundaries: nineteen invariants run as probes against the harness under test, each checked by mutation (the boundary removed, the invariant broken, the boundary restored; Fig. 4b measures all nineteen on every variant). The second tests readings: 107 verbatim quotations that can still be read wrongly, in eight classes, with unsafe errors (those that would let a false claim through) counted apart. The split into 23 development and 84 held-out cases was fixed before the suite first ran, and four blind rounds were each written before the rules that first read them (Fig. 4d). The third tests research value: a blind temporal benchmark in which the store is split at a cutoff year, passages that may have been written on either side belong to neither side, the rules were written before the first run (evals/gold/research_value.yaml) and every read path is bounded in SQL, so that leakage is counted rather than assumed; graph rankings are compared with popularity, chance and two other link predictors, and the pipeline's own predictions are judged by later books.

The other suites measure each capability in the way its task allows: constructed data where an answer can be constructed (planted drug pairs, artificial manuscript traditions, a lost work quoted by four books), a development set where rules need examples, and results the literature already holds where the real corpus can be checked (the taboo renamings of Supplementary Fig. 5, the Yunqi chapters of the Suwen, the first gloss of 几几). Development and constructed sets were written with the rules they test; they are regression tests of intended behaviour, and only held-out or real-corpus results estimate generalization (Supplementary Fig. 2). The two held-out sets of the V2 layer (entities, ancient phenotypes) were annotated before the rules ran on them. Everything but the full-corpus suites reruns on the demonstration corpus in under a minute (`taochronos eval`); the full-corpus values, which need a store that is not distributed, are transcribed from docs/evals.md and marked. Supplementary Table 1 lists every suite.

Supplementary Table 1 | The forty evaluation suites: question, data and split, metrics, result and where the paper reports them (`tables/S1_evaluation_design.csv`)

Layer	Suite	Question	Data and split	Metrics	Result	Reported in
Boundaries	invariants	Do the harness's boundaries hold — scope on every read path, semantic support at commit, objections a model cannot close, AfterToolCall denials, isolation of model code, atomic batches, G7 for experts only, evidence bounds, G8 by stance and level, read-only agent views, integrity, retraction, publication on current versions, acceptance ≠ execution?	19 probes (the audit's counterexamples P01–P10 and nine boundaries added with them) run against the harness under test; six boundaries removed in turn (mutation)	held / exercised / broken	19/19 held (P01–P10 and nine boundaries added with them); each of six boundaries removed on purpose (scope guard, result firewall, semantic support hook, reducer authority, evidence bounds, publish guard) breaks its invariant	Fig. 4b; Supplementary Table 14
Readings	interpretive	Are true quotes read right where the reading is hard — inverted relations, a prohibition beside a prescription, consequences, reported and rejected views, omitted conditions, mixed witnesses, disputed dates, textual cruxes?	107 verbatim cases in 8 classes (gold/interpretive.yaml); 23 development, 84 held-out, split fixed before the first run; 4 blind rounds	accuracy and unsafe errors, per class and per split (dev / held-out)	held-out 82/84 (0.976), 2 unsafe (大黄牡丹汤 read as an indication; 里水, a crux the harvest does not find); dev 23/23. First blind round 43/53 (0.811); round 3, before the crux registry, 7/10 with 3 unsafe cruxes; see below	Fig. 4c,d; Supplementary Table 14
Research value	research_value	On the full corpus split at a cutoff year: do	full corpus split at 1127 and 1368: 230 topic–cutoff pairs read, 189 with	hits and precision@k, mean average	precision@10 0.168 (popularity 0.134; chance 0.054);	Fig. 5; Supplementary Fig. 3;

Layer	Suite	Question	Data and split	Metrics	Result	Reported in
		the graph rankings D4 uses anticipate the remedies later books link to a disease, better than popularity and chance? Do the pipeline's own predictions, made on the past alone, come true?	a link that came true, 29,467 candidates; the pipeline run on the past view at 1368 for 2 development and 4 held-out topics; replicated on the open corpus (Supplementary Fig. 3)	precision, paired sign tests over topics; predictions held (strict / loose), per split and review status; leakage	average precision above popularity in 149 of 189 pairs ($P = 2.4 \times 10^{-16}$); held-out predictions: discoveries 7/19; open corpus: above popularity 24/6 ($P = 0.0014$), not separable from chance 13/17 ($P = 0.58$)	Supplementary Table 14
Verification	philology	Are contested readings kept with probabilities and an uncertain mass? Are variant spellings normalised for matching?	7 contested readings (gold/philology.yaml)	pass rate	7/7	—
Verification	claims	Does extraction find the right hyperedges (relation + key arguments + negations)?	42 gold hyperedges (gold/claims.yaml), development	precision · recall · F1 · recall by relation · provenance pass rate	P 1.0 · R 0.976 · F1 0.988 (miss: the second 湿痹 sign definition). Before the general PROGNOSIS rule (则愈 / 自愈 / 欲解 / 可治 … / 者死 / 难治 / 不治 …): P 0.976 · R 0.952 · F1 0.964, prognosis recall 0 — this gold is dev	Supplementary Table 14
Verification	provenance	Are all committed claims and evidence verbatim? How deep is the Claim → Pixel chain?	every claim and evidence record a demonstration run commits	verbatim rates · chain depth · level availability	100% verbatim claims and evidence; mean chain depth 7.8; page/line/region/image unavailable (no images in the demo)	Supplementary Table 14
Verification	hallucination	Are fabricated, one-character-perturbed and mis-positioned quotes caught, without flagging real quotes or variant spellings?	60 random passages × 4 probes: verbatim, one character replaced, random characters, shifted span	detection rates · verbatim recall	fabricated 100% · perturbed 100% · wrong span 100% detected; verbatim recall 100%; variant spelling accepted	Fig. 4a; Supplementary Table 14
Verification	contradiction	Contradiction vs conditional vs apparent vs support vs unrelated (including differential diagnosis and transcription)	12 passage pairs (gold/contradictions.yaml)	accuracy · macro-F1 · confusion	12/12	—
Verification	lineage	Formula derivation, citation, transcription, rephrasing, inheritance, opposition	typed links of the demonstration corpus (gold/lineage.yaml)	precision (judged) · recall · unjudged predictions	P 0.867 · R 0.929 — false positives 逍遥散 ← 苓桂术甘汤 and 补阳还五汤 ← 透脓散 (convergent compositions); miss: 王清任's implicit opposition	—
Verification	temporal	Period constraints, earliest attestations (through variant spellings and concepts that predate their names), period leakage	6 period constraints, 6 earliest attestations (gold/temporal.yaml)	accuracies · leakage count	constraints 6/6 · earliest 6/6 · leakage 0	Supplementary Table 14
Verification	anachronism	Are equations of historical and modern concepts blocked, careful phrasings allowed, mappings typed conservatively, agent-asserted equivalence refused?	10 statements, 8 mappings, agent-asserted equivalences (gold/anachronism.yaml)	statement accuracy · block P/R · mapping accuracy	statements 10/10 · mappings 8/8 · agent equivalence blocked	Supplementary Table 14
Verification	recovery	After crashes at several points, does resume reach the identical scientific state?	crashes at 4 points of a run; reruns; parallel against sequential runs	success rate · replay consistency	4/4 crash points; deterministic rerun; parallel = sequential; replay consistent	—

Layer	Suite	Question	Data and split	Metrics	Result	Reported in
		Are reruns and parallel runs identical?				
Research value	time_machine	Research on texts before a cutoff (a separate harness, nothing later indexed); are its structured predictions borne out by later texts?	the demonstration corpus split at 1368 (a separate harness indexes the past only)	decidable predictions · accuracy (survived vs other) · leakage · link-prediction baselines	a passage is on one side of the cutoff only when its whole date range is (straddling texts are in neither the past corpus nor the checking future; none at 1368); 3 decidable predictions, 2 held (taboo rename persisted; lost association stayed absent); failed: the 肾气丸-family core predicted from two members — 张景岳's 左/右归丸 dropped 泽泻·牡丹皮·茯苓	—
Research value	source_rediscovery	Hold out a cited book: is it re-inferred as a missing source, dated before its earliest citer?	3 cited books held out (gold/rediscovery.yaml)	recall@3 · date-bound consistency	3/3 re-inferred with consistent date bounds	—
Verification	ablations	Which components carry the results?	demonstration corpus: 6 terms, 12 passage pairs, the agents' context, runs without the Skeptic	earliest accuracy without routes · contradiction accuracy without sense/philology · G5 inflation without clusters · Context-OS token reduction · false survivals without the Skeptic	earliest accuracy: all 1.0 · no sense route 0.67 · BM25 only 0.67; contradiction: no sense resolution 0.83 · no philology 0.92; 1 hypothesis would pass G5 only by counting dependent books; Context OS −94% tokens; without the Skeptic 1 rejected hypothesis survives and no revisions happen	Supplementary Fig. 2c
Deep discovery	collation	Are the variants of a known tradition found, and its stemma, groups and contamination recovered?	5 artificial traditions copied from the demo corpus down a known stemma (two branches; one witness copied from both)	unit recall / precision · split recall · Robinson–Foulds · group precision · contamination P/R	variant units: recall 0.998 · precision 0.999; the true split recovered in every trial (RF 0.33: neighbour-joining resolves the true polytomy — three copies made independently from one exemplar — into an extra split); top-3 groups all genuine (all groups 0.73); contamination P 1.0 · R 1.0	Supplementary Table 14
Deep discovery	reuse	Is each reuse typed right (直接引用 … 套语相似), and are real reuses found?	33 author-constructed pairs (gold/reuse.yaml), all eight types and unrelated pairs	accuracy · macro-F1 · mode accuracy · detection P/R/F1 · pipeline recall	accuracy 0.97 · macro-F1 0.965 · mode 0.97; detection P 1.0 · R 0.96 · F1 0.98; the demo corpus's gold transcriptions and rephrasings found by study.reuse: 1.0	Supplementary Table 14
Deep discovery	stratigraphy	Are the layers of composite texts found, with their boundaries, and are texts attributed to their authors?	5 composites of 16 000 characters with blocks rewritten in a second author's function characters; the real 素问 and 伤寒论	layer accuracy · change-point and boundary P/R · attribution accuracy · real checks (素问 运气七篇, 叔和 chapters)	layer accuracy 0.955; boundaries R 0.90 · P 0.93; change points R 0.37 · P 1.0; attribution 0.89 · real: the 运气七篇 7/7 in the minor layer, both of their boundaries found, 3 of the 4 chapters flagged by late vocabulary are 运气 chapters, 辨脉法 and 平脉法 nearest the 脉经	Supplementary Table 14
Deep discovery	cases	Are case records cut into cases and visits, and are their fields read right?	6 constructed cases in the two manners of the literature — visits under a heading, narratives (gold/cases.yaml); the two transcriptions of 吴鞠通医案	segmentation · per-field accuracy · response and outcome accuracy · agreement of two transcriptions	segmentation, fields, responses, outcomes 1.0 on the development set · real: case counts agree 0.98, outcomes 0.92	—
Deep discovery	argument	Are the steps of reasoning found and typed?	18 author-constructed passages (gold/argument.yaml); the 脉经's two transcriptions against 温病条辨	labelled and unlabelled edge P/R/F1 · clause segmentation · same-work vs other-work divergence	edges F1 1.0 on the development set · real: JSD 0.020 between the transcriptions of one work, 0.091 against another	—
Deep discovery	senses	Is a planted shift of senses dated, and a	synthetic occurrences: one sense rises from 15 % to 75 % in the year	change-year error · detection · labelling	change year within 2.3 years; detected 1.0; labelling 0.93; the	—

Layer	Suite	Question	Data and split	Metrics	Result	Reported in
		sense no cue names found?	1000, an uncued sense appears from 1300	accuracy · hidden sense found · curated exemplars	uncued sense found 1.0 (precision 1.0); curated exemplars of 消渴 6/8	
Deep discovery	fragments	Is a lost text rebuilt from its quotations without taking in other sources' words?	a constructed lost work (集古方, held out) quoted by four books in the manners of the 外台秘要, the 医心方, the 证类本草 and a late paraphrase, beside other sources (gold/fragments/); the real 千金 and 肘后	precision · coverage · decoy leaks · volume accuracy · real verification (千金, 肘后)	precision 0.86 · coverage 1.0 · no other source's words taken in · volumes 1.0 · real: 千金 precision 0.32 · coverage 0.12; 0.876 of the 肘后's quotations not in the extant text	Supplementary Fig. 2a; Supplementary Table 14
Task specification	punctuation	Are the marks of 白文 placed where editors place them, with every character kept?	the store's punctuated texts; 55 works held out of training (104 k characters); 4 Siku 白文 works against their 笈成 punctuation	boundary / sentence F1 · mark-type accuracy · preservation rate · rule baseline · held-out works · Siku 白文	held-out boundary F1 0.767 (P 0.725 · R 0.814; rules 0.235), typed sentence F1 0.543, mark types 0.805; 白文 boundary F1 0.780 (rules 0.303); characters preserved 1.0	Supplementary Fig. 2b; Supplementary Table 14
Task specification	commentaries	Are the commentaries on a clause found in every layout, attributed and compared?	a constructed classic with commentaries in four layouts (gold/commentaries/); the real 太阳之为病	unit P/R/F1 · text F1 · relation F1 · quotation recall · real: known commentators and relations	units F1 1.0 · text F1 0.81 · relations F1 1.0 · quotations 1.0 · real: 25 commentaries, known commentators 11/11, 张卿子 照录 成无己 and 喻昌 承袭 方有执 found	—
Task specification	disputes	Are named views read as rejected or endorsed, and quoted words kept apart?	22 author-constructed sentences (gold/disputes.yaml)	accuracy · reject / endorse P/R/F1 · real: known controversies	accuracy 0.86 · reject P 1.0 R 0.78 · endorse P 0.8 R 1.0 · real: 3/3 known controversies (景岳 on 丹溪's 相火, 越人's 左肾右命门 …)	—
Task specification	variant_impact	Is each variant reading classed by what it changes (dose, negation, drug, finding …)?	30 readings (gold/variant_impact.yaml)	accuracy · macro-F1 · high-impact P/R/F1 · real: the review queue of the 伤寒论	accuracy 0.97 · macro-F1 0.97 · high impact P 1.0 R 0.94 · real: the 伤寒论's 288 units, the top 10 of its review queue all high-impact	—
Task specification	clause	Are the parts of a clause (condition … prognosis) found?	16 clauses of the 伤寒论 and 金匮 (gold/clause.yaml)	piece accuracy · macro-F1 · per-role F1 · segmentation · real: forms of the 伤寒论	pieces 0.977 · macro-F1 0.92 · real: 40 % of the pieces of the fullest 伤寒论 witness read by no rule (平脉法 dialogues, numbering, notes)	—
Task specification	glosses	Are the glosses of a word found, and only of that word?	16 sentences (gold/glosses.yaml)	accuracy · P/R/F1 · real: the first gloss of 几几	accuracy 1.0 · P/R 1.0 · real: 成无己's 伸颈之貌 the first gloss of 几几	—
Task specification	pairs	Do planted drug pairs come first, and do few chance pairs pass Benjamini–Hochberg?	600 synthetic compositions with three planted pairs; the store's 66 196 compositions	planted top-3 · false significant · real: textbook pairs · 方证 links of 桂枝汤 and 麻黄汤	planted top-3 1.0 · false significant 0.0 · real: 7/7 textbook pairs significant with lift > 1.5 (桂枝—芍药 2527 compositions, lift 2.05; 麻黄—杏仁 5.08 …)	Supplementary Fig. 5d; Supplementary Table 14
Task specification	schools	Are a school's planted words found, and do the textbooks' doctrines come out as its own?	5 synthetic 温病 physicians and 16 other authors; the store's schools with the textbooks' doctrines (gold/schools.yaml, 44)	planted top-2 and significant · common words listed · citation shares · real: doctrine recall and attribution	planted words top-2 and significant, no common word listed · real: doctrine recall 0.73 (z ≥ 1.96), attribution 0.64	—
Task specification	names	Is the change of a drug's preferred name dated, and its names equated and completed?	20 synthetic works (薯蓣 → 山药); the dated taboo renames (gold/names.yaml)	switch interval · nested names apart · equations · candidates · real: dated taboo renames	switch interval, nested names, equation, candidate 山芋 all right · real: 2/2 (薯蓣→山药 in 1000–1200, 玄参→元参 in 1600–1800)	Supplementary Fig. 5a,b
Task specification	chain	Does each witness of a passage hang from the witness it was copied from?	6 artificial traditions of 12 witnesses; with each copy regularising spellings on its own	witnesses found · parent accuracy against the most-similar baseline, with and without convergent spellings	parent accuracy 1.0 (most-similar baseline 0.97); with convergent spellings 0.94 (baseline 0.87); witnesses found 0.96–1.0	—
V2	entities	Are the entities of a sentence found and typed (24 types)?	41 dev sentences, 160 entities (gold/ner.yaml); 25 held-out sentences, 117 entities, annotated after the rules were written (gold/ner_heldout.yaml)	precision · recall · F1	dev F1 0.991 · held-out F1 0.776 (P 0.783 · R 0.769)	Supplementary Fig. 2a; Supplementary Table 14
V2	places	Are physicians placed from their signatures,	15 signatures and 5 statements of native place in the forms of real prefaces and biographies, with traps	precision · recall	P/R 1.0 · 84.6 % of physicians placed · real: 291 of 761 works placed, curated places confirmed 4/4, 湿温 in	—

Layer	Suite	Question	Data and split	Metrics	Result	Reported in
		prefaces and biographies?	(gold/places.yaml); the full corpus		清 江南 49 % vs 26 % (q 0.108, not significant)	
V2	translation	Does a translation keep the original layer and add no present-day diagnosis?	24 presentism cases (equations, retrospective diagnoses, modern names a translation adds, comparisons that must pass); the demo corpus	guard accuracy · layer preserved	guard 24/24 · original layer unchanged 24/24 · doublets glossed by their commentators 3/3	—
V2	layout	Are page images read in order, with notes, margins and OCR errors located?	60 synthetic pages (vertical, 双行小注, 版心, 书眉; noisy and character-level variants); four OCR formats; 12 OCR texts with one error each	reading-order accuracy · CER · notes F1	reading order exact 0.983 · CER 0.0005 · notes F1 0.999 · formats identical · errors located 12/12	—
V2	bridge	Are the findings of a passage projected into HPO with typed relations?	20 dev sentences; 12 held-out passages drawn at random from 65 clinical works of the full corpus, annotated before the reader ran on them; 18 canonical descriptions and 2 negatives (gold/bridge.yaml)	precision · recall · F1	dev F1 0.985 · held-out P 0.938 · R 0.732 · F1 0.822, bounds 1.0 · mapping top-1 1.0, caps 3/3, guard 20/20 · 213 HPO ids valid in hp/releases/2026-09-01	Supplementary Fig. 2a; Supplementary Table 14
V2	audit	Does each statement of an answer get the verdict its evidence supports?	19 claims of every kind and 3 safety texts (gold/audit.yaml)	verdict accuracy · false SUPPORTED	verdicts 1.0 · kinds 1.0 · false SUPPORTED 0 · safety 3/3	Supplementary Table 14
V2	modern	Are modern records typed by design and population, and cross-level claims refused?	10 records to type, 7 claim/evidence pairs for the cross-level guard, the synthetic OMOP database (gold/modern.yaml)	typing accuracy · guard accuracy	typing 10/10 · guard 7/7 · RWD: SQL checks pass, planted association recovered, bad SQL refused	—
V2	workflow	Does the orchestrator plan the skills a question needs and no others?	the registry; 8 questions with the skills they need and must not call (gold/workflow.yaml); one end-to-end run	valid plans · rerun from the workspace	registry valid · plans 8/8 · a rerun reads every step from its workspace	—
Verification	aliases	Is a drug found under its other names (国老 for 甘草, 通草 for 木通 before 960) without losing the name asked, merging two drugs that share a name or reaching outside the years a name meant the drug? (below)		synthetic checks; per profile, split and item type, for off, naive, on, analysis and the ablations: P@10 (and its gain where off answers) · nDCG@10 · AskedRet@10 (mean, min) · ON@10 · ON@50 · XR@50 · Wrong@10 · CandRecall · CandPrecision · SilentMerge · FlagP/FlagR · Leakage · latency p50/p95; resolution accuracy	9 planted items on a corpus written for the eval: every check held (side places and retention at k = 5, 10, 20; no merge; gates; readings; periods; flags; no leakage; off = today's search); P@10 off 0.800 · naive 0.806 · on 0.933; Wrong@10 6 · 7 · 3; on: CandRecall 1.0, SilentMerge 0, FlagR 1.0, leakage 0	—

Supplementary Note 2 | Gates, permissions and authority

Five mechanisms bound what any part of the harness can do, and they are layered so that the failure of one is caught by another (Fig. 4b). (1) Permissions: every tool needs one permission, and an agent holds only those its specification lists (Supplementary Tables 4 and 5); an agent that asks for a tool outside them is refused and the refusal is logged. (2) Hooks: deterministic guards at eight points of the lifecycle (Supplementary Table 6) refuse a tool call or result outside the research's scope, a claim whose quote is not verbatim or contradicts it, an anachronistic equation and a report on unjudged versions. (3) Event authority: every change to the research object is an event, and the reducer refuses an event its principal may not emit; 61 event types, nine patterns of permission (Fig. 1b, Supplementary Table 7). An agent proposes; only validators grade, and only a human sets G7. (4) Gates G0–G8 grade each claim, evidence record and hypothesis on its current version (Supplementary Table 2). (5) The research contract, hashed before the first task, fixes the criteria by which a hypothesis is accepted, needs an expert, stays inconclusive or is rejected, and names the judge of each (Supplementary Table 3): finishing the tasks never accepts a research. In conversation, four core agents run the same specialists as fifteen skills (Supplementary Table 8).

Supplementary Table 2 | Gates G0–G8: what each passes, warns and fails, and who sets it (`tables/S2_gates.csv`)

Gate	Name	Subject	Pass	Warn	Fail	Set by
G0	Source	claim, evidence	the book is in the store with an origin and a licence (score 0.8 when the transcription is unverified)	—	unknown book, or no origin or licence	validator
G1	Transcription	claim, evidence	a verified manual transcription, or OCR confidence ≥ 0.9	an unverified transcription, or OCR confidence 0.7–0.9	OCR confidence < 0.7	validator
G2	Philology	claim, evidence	text collated against an edition, and the span not on a contested reading	uncollated text, or a contested reading whose base reading has p ≥ 0.4	the span rests on a contested reading with p < 0.4	validator
G3	Sense	claim; hypothesis	every term resolved to one sense for its period	a homonym whose most likely sense has p < 0.6	—	validator
G4	Claim provenance	claim, evidence; hypothesis	the quote is verbatim at its span and states the relation	verbatim, but the quote does not clearly state the relation, or states a view the author rejects	the quote is not verbatim or not where it is said to be, or it contradicts the claim	validator
G5	Independent sources	hypothesis	supporting evidence from ≥ 2 independent source families (the contract may ask for more)	one family	no supporting evidence	validator
G6	Falsification	hypothesis	the latest Skeptic review leaves no open critical or major objection	an open major objection	an open critical objection (pending before any review)	validator
G7	Expert	hypothesis	a named expert approves	—	a named expert rejects (pending until then)	human only
G8	Modern evidence	hypothesis (biomedical or bridge claims)	records in people support a biomedical claim; any modern record contextualizes a typed bridge	—	no modern record, or a record contradicts the claim; not applicable to historical hypotheses	validator

Supplementary Table 3 | The criteria of the research contract, each with its judge and the state it leaves when unmet (`tables/S3_contract_criteria.csv`)

Criterion	Requirement	Judge	State when unmet	H1 in Fig. 3 (observed)
min_supporting_evidence	≥ 2	validator:contract	inconclusive	pass: 2 verified supporting record(s)
semantic_support	every supporting claim borne by its quote	validator:contract	inconclusive (a quote contradicts its claim, or its author rejects it); needs_expert (a quote does not clearly state its claim)	pass: every supporting claim is stated by its quote
source_dependence	≥ 2 independent source families, on the current version	validator:contract	inconclusive	pass: G5 pass: 2 independent source cluster(s) (need 2): baoming_ji, jc_o128
falsification_review	a skeptic review	validator:contract	inconclusive	pass: reviewed
unresolved_hard_objections	≤ 0	validator:contract	inconclusive	pass: 0 open hard objection(s)
unresolved_critical	≤ 0	validator:contract	inconclusive	pass: 0 open critical objection(s)
alternatives_addressed	every rival explanation of the discovery type excluded or stated as open	validator:contract	inconclusive	pass: all addressed; open: same_name_other_formula
current_version_gates	G0, G1, G2, G3, G4, G5, G6 on the current version (strict: G4, G5, G6)	validator:contract	inconclusive	pass: all required gates pass on this version
expert_review	an expert's approval	human	needs_expert	pending: G7 pending
falsification	not rejected	validator:contract	rejected	R1 in Fig. 3: status rejected

Supplementary Table 4 | The sixteen agents of a research: role, mode, minimum model tier, permissions, tools, output and budget (`tables/S4_agents.csv`)

Agent	Name	Role	Mode	Model tier (minimum)	Permissions	Tools	Output	Budget (tool calls · model calls)
curator	TaoChronos-Curator	Corpus Curator. Freezes scope, hold-out and	procedure	small	classics:read	classics.list_books, vision.layout, vision.compare	CorpusManifest	10 · 0

Agent	Name	Role	Mode	Model tier (minimum)	Permissions	Tools	Output	Budget (tool calls · model calls)
		coverage into the corpus manifest.						
director	TaoChronos	Research Director. Plans each round and decides which specialists to spawn; never judges evidence alone.	hybrid	frontier	classics:read, kg:read, analysis:run, codemode:exec	classics.list_books, classics.search, kg.claims, analysis.temporal_profile	ResearchPlan	30 · 6
evidence	TaoChronos-Evidence	Evidence Retriever. The Evidence Ledger — verbatim, locatable, independently sourced.	hybrid	medium	classics:read, kg:read	classics.search, classics.get_passage, kg.claims, validation.verify_quote, study.cases, study.trajectories, study.argument, graph.evidence	EvidenceSet	80 · 10
evolution	TaoChronos-Evolution	Evolution Analyst. Concept timelines and formula family trees.	procedure	small	analysis:run	analysis.concept_drift, analysis.formula_evolution, analysis.temporal_profile	AnalysisSet	20 · 0
extractor	TaoChronos-Extractor	Claim Extractor. Passages to claim hyperedges with verbatim spans.	hybrid	medium	classics:read, kg:read	kg.extract_claims, kg.term, classics.get_passage, validation.verify_quote, study.clause, study.entities	ClaimSet	80 · 10
hypothesis	TaoChronos-Hypothesis	Hypothesis Generator. Observations to testable, evidence-bound hypotheses; revisions after review.	hybrid	frontier	classics:read, kg:read, codemode:exec	kg.claims, kg.neighbors, kg.term, classics.search, classics.get_passage, validation.verify_quote	HypothesisSet	60 · 10
lineage	TaoChronos-Lineage	Lineage Analyst. Citation, transcription, rephrasing, formula derivation and opposition edges.	procedure	small	classics:read, analysis:run	classics.lineage, philology.collate, analysis.missing_sources, study.schools, study.chain, study.places	LineageReport	20 · 0
meta_reviewer	TaoChronos-MetaReviewer	Meta-Reviewer. Systemic weaknesses, duplicates and next-round recommendations.	hybrid	frontier	kg:read, classics:read	kg.claims, audit.claim	MetaReview	10 · 4
modern_evidence	TaoChronos-ModernEvidence	Modern Evidence Analyst. The only door between the historical and biomedical spaces (Gate G8).	hybrid	frontier	literature:read, kg:read, classics:read	literature.modern_evidence, kg.term, bridge.map, bridge.reverse, modern.search, modern.safety, rwd.ask	ModernEvidenceSet	30 · 6
ontologist	TaoChronos-Ontologist	Ontologist / Terminology Curator. Typed historical-to-modern mappings, proposed only.	hybrid	frontier	kg:read, kg:propose, classics:read	kg.term, kg.propose_change, bridge.phenotype, bridge.map, bridge.reverse	MappingProposals	40 · 6
pattern_miner	TaoChronos-PatternMiner	Pattern Miner. Tool-first discovery engines D1–D5 and source rediscovery; no model.	procedure	small	analysis:run, kg:read, classics:read	analysis.*, study.pairs, study.network	ObservationSet	60 · 0
philologist	TaoChronos-Philologist	Philologist (校勘). Readings, variants, collation status — never a forced single reading.	hybrid	medium	classics:read	philology.*, classics.get_passage, validation.verify_quote, study.variants, study.stemma, study.layers, study.dating, study.authorship, study.witnesses, study.punctuate, study.commentaries, study.glosses, study.clause, study.entities, study.translate, vision.layout, vision.compare	PhilologyReport	60 · 8
scholar	TaoChronos-Scholar	Source Scholar (泊学). The textual history of the focus formulas, drugs and terms over the whole corpus.	procedure	small	classics:read	study.formula, study.herb, study.term, study.senses, study.concordance, study.taboo, study.metrology, study.commentaries, study.disputes, study.network, study.schools,	SourcesDossier	24 · 0

Agent	Name	Role	Mode	Model tier (minimum)	Permissions	Tools	Output	Budget (tool calls · model calls)
						study.names, study.places, study.translate		
semanticist	TaoChronos-Semanticist	Historical Semanticist. Period-bound sense resolution; guards against homonymy.	hybrid	medium	classics:read, kg:read	kg.term, classics.get_passage, classics.search, study.term, study.senses, study.glosses, study.names, study.entities, study.translate, bridge.phenotype	TermResolutionReport	60 · 8
skeptic	TaoChronos-Skeptic	Skeptic (falsification). Tries to break every hypothesis before anyone believes it.	hybrid	frontier	classics:read, kg:read, analysis:run, codemode:exec	classics.search, classics.get_passage, classics.lineage, kg.claims, philology.assess_passage, validation.verify_quote, analysis.temporal_profile, study.reuse, study.dating, study.argument, study.disputes, audit.claim, graph.evidence	ReviewSet	400 · 12
statistician	TaoChronos-Statistician	Statistician. Significance, multiple-testing control and coverage for every observation; no model.	procedure	small	analysis:run	analysis.link_prediction, analysis.temporal_profile	AnalysisSet	20 · 0

Supplementary Table 5 | The tool mesh: families, the permission each needs and the tools (`tables/S5_tools.csv`)

Family	Permission needed	What the permission allows	Tools	Names
analytics	analysis:run	run the discovery and statistics engines	9	analysis.association_rules, analysis.communities, analysis.concept_drift, analysis.contradictions, analysis.formula_evolution, analysis.link_prediction, analysis.lost_knowledge, analysis.missing_sources, analysis.temporal_profile
bridge	classics:read	read the corpus through the research's scoped view	3	bridge.map, bridge.phenotype, bridge.reverse
knowledge	kg:propose	propose a change to canonical knowledge	1	kg.propose_change (writes)
knowledge	kg:read	read the knowledge graph	4	kg.claims, kg.extract_claims, kg.neighbors, kg.term
literature	literature:read	read modern literature and real-world data	4	literature.modern_evidence, modern.safety, modern.search, rwd.ask
mydata	mydata:read	read the user's own data (made anonymous as it is read, never recorded)	6	mydata.compare, mydata.describe, mydata.list, mydata.prescriptions, mydata.read, mydata.search
philology	classics:read	read the corpus through the research's scoped view	6	philology.assess_passage, philology.collate, philology.inspect_page, philology.normalize, vision.compare, vision.layout
research	codemode:exec	run code in the sandbox	1	research.run_code
retrieval	classics:read	read the corpus through the research's scoped view	5	classics.canon, classics.get_passage, classics.lineage, classics.list_books, classics.search
study	classics:read	read the corpus through the research's scoped view	41	study.argument, study.authorship, study.cards, study.cases, study.chain, study.citations, study.clause, study.commentaries, study.compare, study.concordance, study.dating, study.disputes, study.entities, study.family, study.formula, study.fragments, study.glosses, study.herb, study.incompat, study.layers, study.metrology, study.names, study.network, study.pairs, study.places, study.properties, study.protocol, study.punctuate, study.reading, study.reuse, study.schools, study.senses, study.stemma, study.taboo,

Family	Permission needed	What the permission allows	Tools	Names
				study.term, study.trajectories, study.translate, study.transmission, study.usage, study.variants, study.witnesses
validation	classics:read	read the corpus through the research's scoped view	4	audit.claim, graph.evidence, validation.provenance_chain, validation.verify_quote
validation	validation:run	run the gates (validators)	1	validation.gates

Supplementary Table 6 | The hooks the harness runs and what each refuses (`tables/S6_hooks.csv`)

Hook point	Guard	Runs on	Refuses
BeforeToolCall	scope_guard	every tool call	arguments that name material outside the research's scope (withheld books, the future of a blind cutoff)
AfterToolCall	scope_firewall	every tool result	a result that carries withheld material back to the caller
BeforeClaimCommit	provenance_validator	every claim	a quote that is not verbatim, empty or outside its passage; inconsistent argument spans
BeforeClaimCommit	semantic_support	every claim	a claim its quote contradicts (a true quote with a false relation); frames the rest with speaker, stance, conditions and negations
BeforeClaimCommit	anachronism_guard	every claim	a classical claim that carries a modern concept
AfterHypothesis	anachronism_guard	every hypothesis and revision	a statement that equates a historical concept with a modern category, or breaks a forbidden assumption of the contract
BeforeGraphWrite	anachronism_guard	every mapping into the knowledge graph	a historical-to-modern equivalence asserted by anyone but a human expert
BeforeArtifactPublish	publish_guard	every report that is not a draft	a report without evidence, one that cites missing or retracted objects, or one that presents a hypothesis not judged on its current version

Supplementary Table 7 | Event authority: the principals allowed to emit each event type (● allowed), with the reason where the table records one (`tables/S7_event_authority.csv`)

Event	Agent	Validator	Human	Kernel	Why
ResearchCreated				●	only the harness opens a research contract
ProfileLoaded				●	
CorpusScoped	●	●		●	
RoundStarted				●	
RoundCompleted				●	
StopConditionMet				●	
ResearchCompleted				●	only the harness completes a research
ResearchStopped			●	●	
ContractEvaluated		●		●	the research contract is evaluated by validators
TaskPlanned	●	●		●	
TaskStarted	●	●		●	
TaskCompleted	●	●		●	
TaskFailed	●	●		●	
TaskSkipped	●	●		●	
AgentSpawned				●	
AgentReleased				●	
ModelRouted				●	
LLMCalled	●			●	
ToolCalled	●	●	●	●	
ToolResult	●	●	●	●	
ToolDenied	●	●	●	●	
HookBlocked	●	●	●	●	
BudgetExceeded	●	●	●	●	
BudgetReserved				●	
BudgetSettled				●	
RunManifestRecorded				●	

Event	Agent	Validator	Human	Kernel	Why
PassageAssessed	•	•		•	
TermResolved	•	•		•	
EvidenceRetrieved	•	•		•	
EvidenceRetracted		•	•	•	
ClaimExtracted	•	•		•	
ClaimRejected	•	•		•	
LineageDetected	•	•		•	
ContradictionFound	•	•		•	
ObservationRecorded	•	•		•	
AnalysisRecorded	•	•		•	
WarrantRecorded	•	•		•	
HypothesisGenerated	•	•		•	
CounterEvidenceFound	•	•		•	
HypothesisReviewed	•	•		•	
HypothesisRevised	•	•		•	
HypothesisRejected		•	•	•	
HypothesisScored		•		•	scores are computed by validators
HypothesisRanked		•		•	rankings are computed by validators
HypothesisConfirmed		•		•	confirmation is a validator's re-test on withheld material
GateEvaluated		•		•	gates are graded by validators, never by the work they grade
ObjectionResolutionProposed	•	•		•	
ObjectionClosed		•	•		an objection is closed by re-running its check (validator) or by an expert, never by a model
ChangeProposed	•	•	•	•	
ChangeApproved		•	•		agents may only propose changes; a validator or human must decide
ChangeRejected		•	•		agents may only propose changes; a validator or human must decide
ExpertApproved			•		only a human expert may approve or reject a hypothesis (Gate G7)
ExpertRejected			•		only a human expert may approve or reject a hypothesis (Gate G7)
ExpertSteered			•		only a human expert steers a research
DecisionRecorded	•	•	•	•	
MemoryWritten		•	•	•	
BranchForked				•	
BranchMerged				•	
CheckpointCreated				•	
ArtifactPublished		•		•	publication is a validator's act after the contract is evaluated
TransactionCommitted				•	

Supplementary Table 8 | In conversation: the four core agents, the skills S01–S15 they run and the specialists they delegate to (`tables/S8_core_agents_skills.csv`)

Core agent	Skills	Specialists it delegates to	Permissions	Rules
古籍学者 Classical TCM Scholar	S01 来源与版本 Source & Provenance; S02 影像与版面 Document Vision; S03 句读·训诂 Philology; S04 条文结构 Structure & Clause; S05 校勘 Collation; S06 互文与传承 Intertext; S07 知识抽取 Knowledge Extraction; S08 名称规范 TCM Normalization; S09 源流演变 Historical Evolution	curator, philologist, extractor, semanticist, lineage, scholar, evolution	classics:read, kg:read	the original is never changed; every reading is a layer with its source / absence is only meaningful where the period's coverage is adequate
证据与批评 Evidence & Critic Agent	S15 证据审计 Evidence Audit	evidence, skeptic, statistician	classics:read, kg:read, literature:read, validation:run	a verdict is never raised by a model / SUPPORTED means the checks this system can run pass, not that the claim is true
研究编排者 Research Orchestrator	plan plan; synthesize synthesize	director, meta_reviewer	classics:read, kg:read, analysis:run	rules plan first; a model may add or drop optional skills, never S01 or S15, and never a skill the registry lacks / the answer says what the evidence says and no more; the critic's verdict is shown beside each statement
古今转译 Translational Medicine Agent	S10 古代表型化 Ancient Phenotyping; S11 本体桥接 Ontology Bridge; S12 古今映射 Clinical Mapping; S13 现代证据 Modern Evidence; S14 真实世界验证 RWD Validation	ontologist, modern_evidence, hypothesis	classics:read, kg:read, literature:read	phenotypic similarity, never retrospective diagnosis; no ancient name is equated with a modern disease / animal, cell and computational work is not efficacy in people; no clinical or dosage advice

Supplementary Note 3 | The end-to-end research case

The question '肾气丸的组成与主治如何随时代演变？ was put to the harness on the open canon corpus with `taochronos research "肾气丸的组成与主治如何随时代演变？" --profile web-canon --focus 肾气丸,六味地黄丸 --forbid 消渴=糖尿病`, offline: the rules ran alone and no model was called. The run scoped 273 books and 1,620 passages, planned 30 tasks in three rounds, called 143 tools (none refused), extracted 935 claims, recorded 80 evidence and 13 counter-evidence records and 25 hypotheses, and stopped after round 2, the last the contract allowed, which added no new evidence. It is deterministic: a second run writes the same report. The contract's verdict: four hypotheses wait for an expert — two (H1, H2) meet every criterion a machine can check, and two (H3, H4) rest in part on a quotation that does not clearly state its claim, which the contract leaves to an expert — nine are inconclusive and one (R1) is rejected. Supplementary Data 1 is the whole Discovery Report the run wrote, in Chinese, as a user of the harness receives it; Supplementary Data 2 is its event log, from which the state replays; Supplementary Tables 9 and 10 list every hypothesis and objection. Reading each surviving hypothesis against its evidence found six failure modes (Supplementary Fig. 1), the largest being that the independence check merged 36 of the 46 books that supplied evidence into one source family.

Supplementary Table 9 | Every hypothesis of the end-to-end run (Fig. 3), with its gates, unmet criteria, objections and testable prediction (`tables/S9_e2e_hypotheses.csv`)

Label	Id	Kind	Generation	Parent	Status	Verdict	Gates G0–G8	Criteria unmet	Objectio (closed)
H1	hyp_c19dcbbde26a	association_rule	0		survived	needs_expert	G0:pass G1:warn G2:warn G3:pass G4:pass G5:pass G6:pass G7:pending G8:not_applicable	expert_review	0 (0)
H2	hyp_81b0e8188593	concept_drift	1	hyp_1ebef0b387ae	survived	needs_expert	G0:pass G1:warn G2:warn G3:warn G4:pass G5:pass G6:pass G7:pending G8:not_applicable	expert_review	3 (2)
H3	hyp_44606b0576bb	concept_drift	1	hyp_8d1ea31b9d9a	survived	needs_expert	G0:pass G1:warn G2:warn G3:warn G4:pass G5:pass G6:pass	semantic_support, expert_review	3 (2)

Label	Id	Kind	Generation	Parent	Status	Verdict	Gates G0–G8	Criteria unmet	Objectic (closed)
							G7:pending G8:not_applicable		
H4	hyp_505a55a4cb6b	concept_drift	1	hyp_6050f468fdcf	survived	needs_expert	G0:pass G1:warn G2:warn G3:warn G4:pass G5:pass G6:pass G7:pending G8:not_applicable	semantic_support, expert_review	4 (3)
H5	hyp_748f1bdebc6e	association_rule	1	hyp_9dcce49d684f	survived	inconclusive	G0:pass G1:warn G2:warn G3:pass G4:pass G5:warn G6:pass G7:pending G8:not_applicable	source_dependence, current_version_gates, expert_review	4 (4)
H6	hyp_e2ebd991ae10	lost_source	0		survived	inconclusive	G0:pass G1:warn G2:warn G3:pass G4:pass G5:warn G6:pass G7:pending G8:not_applicable	source_dependence, current_version_gates, expert_review	2 (0)
H7	hyp_266b1439da88	lost_source	1	hyp_00ee5a11ff84	survived	inconclusive	G0:pass G1:warn G2:warn G3:pass G4:pass G5:warn G6:pass G7:pending G8:not_applicable	source_dependence, current_version_gates, expert_review	4 (2)
H8	hyp_f81fd13de194	lost_knowledge	1	hyp_6ffa899e3703	survived	inconclusive	G0:pass G1:warn G2:warn G3:pass G4:pass G5:warn G6:pass G7:pending G8:not_applicable	min_supporting_evidence, source_dependence, current_version_gates, expert_review	8 (8)

Label	Id	Kind	Generation	Parent	Status	Verdict	Gates G0–G8	Criteria unmet	Objectic (closed)
H9	hyp_97cca9a0a038	contradiction	1	hyp_b8a726ad7aef	survived	inconclusive	G0:pass G1:warn G2:warn G3:warn G4:pass G5:warn G6:pass G7:pending G8:not_applicable	semantic_support, source_dependence, current_version_gates, expert_review	8 (4)
H10	hyp_18f4717f2182	contradiction	1	hyp_01f85940a40d	survived	inconclusive	G0:pass G1:warn G2:warn G3:warn G4:pass G5:warn G6:pass G7:pending G8:not_applicable	semantic_support, source_dependence, current_version_gates, expert_review	8 (4)
H11	hyp_eb8e94beb10d	contradiction	1	hyp_b7aadb438d6b	survived	inconclusive	G0:pass G1:warn G2:warn G3:warn G4:pass G5:warn G6:pass G7:pending G8:not_applicable	semantic_support, source_dependence, current_version_gates, expert_review	8 (4)
H12	hyp_dfcf54b7ed92	hidden_association	1	hyp_93968335aee1	survived	inconclusive	G0:pass G1:warn G2:warn G3:warn G4:pass G5:warn G6:pass G7:pending G8:not_applicable	source_dependence, current_version_gates, expert_review	4 (4)
H13	hyp_675ae4dcc3a3	hidden_association	1	hyp_95577d14638d	survived	inconclusive	G0:pass G1:warn G2:warn G3:pass G4:pass G5:warn G6:pass G7:pending G8:not_applicable	source_dependence, current_version_gates, expert_review	6 (6)
R1	hyp_cd02eca5e08b	lost_knowledge	0		rejected	rejected	G0:pass G1:warn G2:warn G3:pass G4:pass G5:warn G6:fail G7:pending G8:not_applicable		5 (0)
(superseded)	hyp_00ee5a11ff84	lost_source	0		superseded		G0:· G1:· G2:· G3:· G4:· G5:· G6:· G7:pending G8:·		2 (0)

Label	Id	Kind	Generation	Parent	Status	Verdict	Gates G0–G8	Criteria unmet	Objectic (closed)
(superseded)	hyp_01f85940a40d	contradiction	0		superseded		G0:· G1:· G2:· G3:· G4:· G5:· G6:· G7:pending G8:·		4 (0)
(superseded)	hyp_1ebef0b387ae	concept_drift	0		superseded		G0:· G1:· G2:· G3:· G4:· G5:· G6:· G7:pending G8:·		3 (0)
(superseded)	hyp_6050f468fdcf	concept_drift	0		superseded		G0:· G1:· G2:· G3:· G4:· G5:· G6:· G7:pending G8:·		4 (0)
(superseded)	hyp_6ffa899e3703	lost_knowledge	0		superseded		G0:· G1:· G2:· G3:· G4:· G5:· G6:· G7:pending G8:·		4 (0)
(superseded)	hyp_8d1ea31b9d9a	concept_drift	0		superseded		G0:· G1:· G2:· G3:· G4:· G5:· G6:· G7:pending G8:·		3 (0)
(superseded)	hyp_93968335aee1	hidden_association	0		superseded		G0:· G1:· G2:· G3:· G4:· G5:· G6:· G7:pending G8:·		2 (0)
(superseded)	hyp_95577d14638d	hidden_association	0		superseded		G0:· G1:· G2:· G3:· G4:· G5:· G6:· G7:pending G8:·		3 (0)
(superseded)	hyp_9dcce49d684f	association_rule	0		superseded		G0:· G1:· G2:· G3:· G4:· G5:· G6:· G7:pending G8:·		2 (0)
(superseded)	hyp_b7aadb438d6b	contradiction	0		superseded		G0:· G1:· G2:· G3:· G4:· G5:· G6:· G7:pending G8:·		4 (0)

Label	Id	Kind	Generation	Parent	Status	Verdict	Gates G0–G8	Criteria unmet	Objectic (closed)
(superseded)	hyp_b8a726ad7aef	contradiction	0		superseded		G0:· G1:· G2:· G3:· G4:· G5:· G6:· G7:pending G8:·		4 (0)

Supplementary Table 10 | The Skeptic's 102 objections in the end-to-end run, by check (each objection is in the CSV file) (`tables/S10_e2e_objections.csv`)

Check	Raised	Closed after narrowing	Open: major	Open: minor	Open: critical
transcription_dependence	28	14	8	6	0
single_source	27	8	5	14	0
homonym	11	7	4	0	0
school	9	0	0	9	0
statistical	9	5	4	0	0
sample_size	6	3	3	0	0
partial_persistence	4	2	2	0	0
earliest_source	4	2	2	0	0
counterexample	3	2	1	0	0
restated_after	1	0	0	0	1

Supplementary Note 4 | Prediction lists

Two lists record what the harness predicted before it could see the outcome, on the open canon corpus. Supplementary Table 11 lists the 82 predictions of the whole pipeline run on the past view at 1368 for two development topics (消渴, 中风) and four held-out topics (胸痹, 霍乱, 虚劳, 痰饮), each with its outcome in the later books (32 could be judged by the benchmark; formula evolutions, contradictions and lost books are not). No passage outside the past view was read (leakage 0). Supplementary Table 12 lists, for each of the 51 topic–cutoff pairs in which a link came true (109 links among 10,115 candidates), the average precision of three rankers and the true links Adamic–Adar placed in its top 20. The full-corpus runs of Fig. 5 (189 pairs, 1,314 links) need the store that is not distributed; their aggregates are in docs/evals.md.

Supplementary Table 11 | The predictions of the pipeline run on the past view at 1368 on the open corpus, and their outcomes (`tables/S11_predictions_open_corpus.csv`)

Topic	Split	Hypothesis	Kind	Review status	Prediction	Subject	Object	Held (strict)	Held (loose)	Later books	Note
disease: 消渴	dev	hyp_5608172ffb13	hidden_association	rejected	link_appears	symptom:咳	formula:五苓散	no	no	1	already written before the cutoff
disease: 消渴	dev	hyp_af6d5b43f7af	lost_source	survived	link_appears	work:wupu_bencao		not judged	not judged		not judged by this benchmark
disease: 消渴	dev	hyp_a80a336128c2	concept_drift	survived	sense_shift	disease:消渴	消渴#病·多饮多尿	yes	yes		
disease: 消渴	dev	hyp_4ad17f25a523	lost_knowledge	survived	absent_after	formula:五苓散	disease:消渴	no	no	5	
disease: 消渴	dev	hyp_576a55a1ac17	formula_evolution	survived	herb_retained	herb:半夏		not judged	not judged		not judged by this benchmark
disease: 消渴	dev	hyp_dc5e204b197e	lost_source	survived	link_appears	work:jingxin_lu		not judged	not judged		not judged by this benchmark
disease: 消渴	dev	hyp_9f1c3f0ed371	formula_evolution	survived	herb_retained	herb:猪苓,herb:白术,herb:茯苓		not judged	not judged		not judged by this benchmark
disease: 消渴	dev	hyp_823d383ae14e	contradiction	survived	resolution	disease:伤寒	contradiction	not judged	not judged		not judged by this benchmark
disease: 消渴	dev	hyp_7027c95205fd	association_rule	survived	cooccurrence_after	symptom:不渴,symptom:头痛	formula:越婢汤	no	no		
disease: 消渴	dev	hyp_4f0e6de21e7e	association_rule	survived	cooccurrence_after	symptom:头痛,symptom:无大热	formula:越婢汤	not judged	not judged		
disease: 消渴	dev	hyp_b4c92c32c385	contradiction	survived	resolution	disease:伤寒	contradiction	not judged	not judged		not judged by this benchmark
disease: 消渴	dev	hyp_27c77e0fa40a	formula_evolution	survived	herb_retained	formula:苓桂术甘汤,formula:细辛散		not judged	not judged		not judged by this benchmark
disease: 消渴	dev	hyp_514347a5ab71	hidden_association	survived	link_appears	symptom:小便不利	formula:麻黄汤	no	no	0	
disease: 消渴	dev	hyp_08b2269de51a	lost_knowledge	survived	absent_after	etiology:服石	disease:消渴	yes	no	0	

Topic	Split	Hypothesis	Kind	Review status	Prediction	Subject	Object	Held (strict)	Held (loose)	Later books	Note
disease:中风	dev	hyp_b556c451a515	lost_knowledge	rejected	absent_after	etiology:风	disease:中风	no	no	12	
disease:中风	dev	hyp_cd4f103599a3	hidden_association	rejected	link_appears	symptom:不恶寒	formula:桂枝汤	no	no	0	
disease:中风	dev	hyp_057d771c5611	lost_source	survived	link_appears	work:主治秘要		not judged	not judged		not judged by this benchmark
disease:中风	dev	hyp_3495600970d8	contradiction	survived	resolution	pathogenesis:里寒	contradiction	not judged	not judged		not judged by this benchmark
disease:中风	dev	hyp_194ef9aa2f95	contradiction	survived	resolution	disease:中风	conditional	not judged	not judged		not judged by this benchmark
disease:中风	dev	hyp_d1a0a9a4aa27	hidden_association	survived	link_appears	symptom:喘	formula:小柴胡汤	no	no	0	
disease:中风	dev	hyp_6a5c759c767b	association_rule	survived	cooccurrence_after	symptom:下利,symptom:半身不遂	formula:姜附汤	not judged	not judged		
disease:中风	dev	hyp_89303484a9d1	formula_evolution	survived	herb_retained	herb:大枣,herb:甘草,herb:生姜,herb:芍药		not judged	not judged		not judged by this benchmark
disease:中风	dev	hyp_51f54da65656	lost_source	survived	link_appears	work:玉函		not judged	not judged		not judged by this benchmark
disease:中风	dev	hyp_dca3de12f279	association_rule	survived	cooccurrence_after	pulse:微,symptom:头痛	formula:鸡屎白散	no	no		
disease:中风	dev	hyp_d83e15e95ac8	lost_knowledge	survived	absent_after	formula:五苓散	disease:中风	no	no	6	
disease:中风	dev	hyp_e0c76e58ffd1	formula_evolution	survived	herb_retained	herb:杏仁,herb:甘草,herb:石膏,herb:麻黄		not judged	not judged		not judged by this benchmark
disease:中风	dev	hyp_5ed6252ce76e	formula_evolution	survived	herb_retained	herb:甘草,herb:葛根,herb:黄芩		not judged	not judged		not judged by this benchmark
disease:中风	dev	hyp_14b5ea5d7a81	concept_drift	survived	sense_shift	disease:中风		not judged	not judged		
disease:胸痹	heldout	hyp_4af9e0e3e79c	lost_source	survived	link_appears	work:gujin_luyan		not judged	not judged		not judged by this benchmark
disease:胸痹	heldout	hyp_5ebe75623097	formula_evolution	survived	herb_retained	herb:栝楼实,herb:薤白		not judged	not judged		not judged by this benchmark
disease:胸痹	heldout	hyp_a43b2b13831c	formula_evolution	survived	herb_retained	herb:甘草,herb:白术,herb:茯苓		not judged	not judged		not judged by this benchmark
disease:胸痹	heldout	hyp_7b3550c17534	association_rule	survived	cooccurrence_after	pulse:微	formula:附子汤	no	no		
disease:胸痹	heldout	hyp_0910282fd418	contradiction	survived	resolution	disease:胸痹	contradiction	not judged	not judged		not judged by this benchmark
disease:胸痹	heldout	hyp_fa95353f8b7c	contradiction	survived	resolution	disease:胸痹	contradiction	not judged	not judged		not judged by this benchmark
disease:胸痹	heldout	hyp_d05ebd4bb9c5	contradiction	survived	resolution	disease:胸痹	contradiction	not judged	not judged		not judged by this benchmark
disease:胸痹	heldout	hyp_664857ebf4f8	contradiction	survived	resolution	disease:胸痹	contradiction	not judged	not judged		not judged by this benchmark
disease:胸痹	heldout	hyp_ae095c1a8d06	association_rule	survived	cooccurrence_after	pulse:微,symptom:拘急	formula:附子汤	no	no		
disease:胸痹	heldout	hyp_a5b35de6aeeb	hidden_association	survived	link_appears	symptom:咳	formula:薤白汤	no	no	0	
disease:胸痹	heldout	hyp_6d7c0046ef32	hidden_association	survived	link_appears	disease:胸痹	herb:栝楼实	no	no	1	
disease:胸痹	heldout	hyp_27916a874d7c	concept_drift	survived	sense_shift	disease:胸痹		not judged	not judged		
disease:霍乱	heldout	hyp_1101b63a75f5	hidden_association	rejected	link_appears	disease:少阴病	formula:五苓散	no	no	0	
disease:霍乱	heldout	hyp_4e38ff74790d	concept_drift	survived	sense_shift	disease:霍乱		not judged	not judged		
disease:霍乱	heldout	hyp_64658eb3b657	hidden_association	rejected	link_appears	symptom:心烦	formula:四逆汤	no	no	0	
disease:霍乱	heldout	hyp_fab01fabcef8	lost_source	survived	link_appears	work:廣志		not judged	not judged		not judged by this benchmark
disease:霍乱	heldout	hyp_01ac93cc8c2c	lost_source	survived	link_appears	work:jinxiao_fang		not judged	not judged		not judged by this benchmark
disease:霍乱	heldout	hyp_24329942a5fb	contradiction	survived	resolution	disease:伤寒	contradiction	not judged	not judged		not judged by this benchmark

Topic	Split	Hypothesis	Kind	Review status	Prediction	Subject	Object	Held (strict)	Held (loose)	Later books	Note
disease:霍乱	heldout	hyp_756adeeaf009	association_rule	survived	cooccurrence_after	pulse:细,symptom:心烦	formula:橘皮汤	not judged	not judged		
disease:霍乱	heldout	hyp_2d929f9cfa0d	association_rule	survived	cooccurrence_after	pulse:弦,symptom:心烦	formula:橘皮汤	not judged	not judged		
disease:霍乱	heldout	hyp_e2d7d528bdc3	lost_knowledge	survived	absent_after	etiology:生冷	disease:霍乱	yes	no	0	
disease:霍乱	heldout	hyp_6a1c6ff4847d	lost_knowledge	survived	absent_after	etiology:湿	disease:霍乱	no	no	3	
disease:霍乱	heldout	hyp_fb3ec7b257f7	formula_evolution	survived	herb_retained	herb:桂心,herb:泽泻,herb:猪苓,herb:白术,herb:茯苓		not judged	not judged		not judged by this benchmark
disease:霍乱	heldout	hyp_5ddb1771ada9	contradiction	survived	resolution	disease:伤寒	contradiction	not judged	not judged		not judged by this benchmark
disease:霍乱	heldout	hyp_0bd7235e7c93	formula_evolution	survived	herb_retained	formula:桂枝汤,formula:桂枝加厚朴杏仁汤,formula:桂枝加芍药汤,formula:桂枝加葛根汤,formula:桂枝加黄芪汤		not judged	not judged		not judged by this benchmark
disease:霍乱	heldout	hyp_6bbe7b001ba1	formula_evolution	survived	herb_retained	herb:桔梗,herb:白术,herb:葛根		not judged	not judged		not judged by this benchmark
disease:虚劳	heldout	hyp_6e2c442d68f4	lost_source	survived	link_appears	work:gujin_luyan		not judged	not judged		not judged by this benchmark
disease:虚劳	heldout	hyp_eb599602bc5b	hidden_association	rejected	link_appears	disease:虚劳	formula:阳旦汤	no	no	0	
disease:虚劳	heldout	hyp_981dfc9b5035	concept_drift	survived	sense_shift	disease:虚劳		not judged	not judged		
disease:虚劳	heldout	hyp_71c8050c5b39	contradiction	survived	resolution	symptom:喘	contradiction	not judged	not judged		not judged by this benchmark
disease:虚劳	heldout	hyp_7f5d2b067426	lost_source	survived	link_appears	work:虚劳脉證第六		not judged	not judged		not judged by this benchmark
disease:虚劳	heldout	hyp_ef9914bc5ca5	hidden_association	rejected	link_appears	disease:中风	formula:桂枝汤	yes	yes	3	already written before the cutoff
disease:虚劳	heldout	hyp_69154ca7fc1d	association_rule	survived	cooccurrence_after	pulse:微,symptom:拘急	formula:附子汤	no	no		
disease:虚劳	heldout	hyp_3d09581465f5	formula_evolution	survived	herb_retained	herb:地黄		not judged	not judged		not judged by this benchmark
disease:虚劳	heldout	hyp_bcd3d8772231	association_rule	survived	cooccurrence_after	symptom:心烦,symptom:鼻衄	formula:小建中汤	not judged	not judged		
disease:虚劳	heldout	hyp_757226e1d312	contradiction	survived	resolution	disease:虚劳	contradiction	not judged	not judged		not judged by this benchmark
disease:虚劳	heldout	hyp_8146f24b4b41	lost_knowledge	survived	absent_after	pathogenesis:痰	disease:虚劳	no	no	3	
disease:虚劳	heldout	hyp_5507d6f5ac8e	formula_evolution	survived	herb_retained	herb:人参,herb:甘草,herb:麦冬		not judged	not judged		not judged by this benchmark
disease:虚劳	heldout	hyp_35e9d86013f7	lost_knowledge	survived	absent_after	formula:桂枝汤	disease:虚劳	yes	yes	0	
disease:虚劳	heldout	hyp_24bee37e9aba	formula_evolution	survived	herb_retained	herb:姜,herb:黄芪		not judged	not judged		not judged by this benchmark
disease:痰饮	heldout	hyp_6e2c442d68f4	lost_source	survived	link_appears	work:gujin_luyan		not judged	not judged		not judged by this benchmark
disease:痰饮	heldout	hyp_5299b6d78bc6	hidden_association	rejected	link_appears	symptom:下利	formula:甘草汤	no	no	0	
disease:痰饮	heldout	hyp_5730b22d8213	lost_source	survived	link_appears	work:jiyan_fang		not judged	not judged		not judged by this benchmark
disease:痰饮	heldout	hyp_a5e57687fd43	hidden_association	rejected	link_appears	symptom:呕吐	formula:泽泻汤	no	no	0	
disease:痰饮	heldout	hyp_40c55f5cfa8a	formula_evolution	survived	herb_retained	herb:厚朴,herb:葛根,herb:陈皮		not judged	not judged		not judged by this benchmark
disease:痰饮	heldout	hyp_de003d6e5453	lost_knowledge	survived	absent_after	etiology:寒	disease:痰饮	no	no	4	
disease:痰饮	heldout	hyp_c4211fadc6e5	lost_knowledge	survived	absent_after	formula:小半夏汤	disease:痰饮	yes	yes	0	
disease:痰饮	heldout	hyp_aebb0187f181	concept_drift	survived	sense_shift	disease:痰饮		not judged	not judged		
disease:痰饮	heldout	hyp_d4166477be17	concept_drift	survived	sense_shift	disease:伤寒	伤寒#广义	no	no		

Topic	Split	Hypothesis	Kind	Review status	Prediction	Subject	Object	Held (strict)	Held (loose)	Later books	Note
disease:痰饮	heldout	hyp_6682c9b596f1	association_rule	survived	cooccurrence_after	pulse:微,symptom:拘急	formula:附子汤	no	no		
disease:痰饮	heldout	hyp_ae8cc119ea3c	association_rule	survived	cooccurrence_after	pulse:微	formula:附子汤	no	no		
disease:痰饮	heldout	hyp_529d6ca54699	formula_evolution	survived	herb_retained	herb:人参,herb:白术		not judged	not judged		not judged by this benchmark
disease:痰饮	heldout	hyp_20f5f4132d38	concept_drift	survived	sense_shift	disease:中风	风#外风	no	no		
disease:痰饮	heldout	hyp_c2c854128b6c	contradiction	survived	resolution	disease:痰饮	contradiction	not judged	not judged		not judged by this benchmark

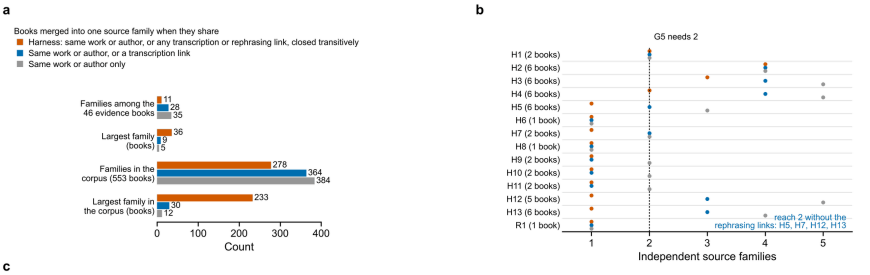
Supplementary Table 12 | Topic–cutoff pairs of the open corpus in which a link came true: average precision of three rankers and the true links Adamic–Adar ranked in its top 20 (`tables/S12_anticipated_links_open_corpus.csv`)

Cutoff	Topic	Past passages	Future passages	Candidates	Came true	AP Adamic–Adar	AP popularity	AP chance	First true link (Adamic–Adar rank)	True links in Adamic–Adar's top 20
1127	disease:中风	763	1325	172	1	0.0667	0.0074	0.0245	15	formula:麻黄汤 (rank 15, 2 later books)
1127	disease:伤寒	1971	3526	231	8	0.1832	0.1967	0.0565	2	herb:人参 (rank 2, 3 later books); herb:麻黄 (rank 4, 3 later books)
1127	disease:痹	1766	1685	226	1	0.0128	0.0045	0.0123	78	
1127	disease:结胸	212	322	48	2	0.0662	0.0431	0.1014	22	
1127	disease:霍乱	372	643	120	1	0.1	0.0084	0.0231	10	formula:四逆汤 (rank 10, 2 later books)
1127	symptom:下利	1145	1765	172	5	0.0486	0.0476	0.0609	22	
1127	symptom:下血	409	586	114	1	0.0118	0.0125	0.0348	85	
1127	symptom:不渴	266	386	35	1	0.0333	0.037	0.1115	30	
1127	symptom:发热	1673	3674	166	4	0.0377	0.026	0.048	39	
1127	symptom:呕吐	857	1758	244	5	0.063	0.015	0.0503	5	formula:五苓散 (rank 5, 5 later books)
1127	symptom:咳	1748	2457	228	2	0.0913	0.0554	0.0295	6	herb:杏仁 (rank 6, 2 later books)
1127	symptom:喘	1485	2272	161	4	0.3181	0.3953	0.0417	1	herb:甘草 (rank 1, 2 later books); herb:人参 (rank 19, 2 later books)
1127	symptom:头痛	984	1805	145	2	0.1349	0.0189	0.0342	4	formula:四逆汤 (rank 4, 4 later books)
1127	symptom:寒热	1742	2447	250	1	0.0059	0.0048	0.0204	169	
1127	symptom:小便不利	570	648	84	1	0.0417	0.0137	0.0896	24	
1127	symptom:心烦	675	724	142	2	0.027	0.0137	0.0496	40	
1127	symptom:拘急	366	453	170	1	0.006	0.0093	0.0273	167	
1127	symptom:无汗	249	756	18	1	0.0588	0.0667	0.1659	17	formula:茵陈汤 (rank 17, 2 later books)
1127	symptom:汗出	2234	2516	207	3	0.0265	0.0118	0.0461	37	
1127	symptom:浊	397	1332	93	1	0.0244	0.0227	0.0728	41	

Cutoff	Topic	Past passages	Future passages	Candidates	Came true	AP Adamic–Adar	AP popularity	AP chance	First true link (Adamic–Adar rank)	True links in Adamic–Adar's top 20
1127	symptom:渴	1687	3531	225	5	0.0272	0.0183	0.0318	67	
1127	symptom:腹满	1010	926	168	4	0.1152	0.0181	0.0778	3	formula:桂枝汤 (rank 3, 2 later books)
1127	symptom:腹痛	976	1826	207	3	0.2778	0.3432	0.0449	3	herb:当归 (rank 3, 2 later books); herb:人参 (rank 7, 2 later books); formula:四逆汤 (rank 14, 2 later books)
1127	symptom:谵语	316	543	34	1	0.0435	0.0588	0.1126	23	
1368	disease:中风	1123	957	262	1	0.0043	0.0068	0.0122	232	
1368	disease:伤寒	2943	2545	306	3	0.155	0.186	0.0247	3	herb:人参 (rank 3, 3 later books)
1368	disease:热病	781	433	57	1	0.0175	0.0256	0.0639	57	
1368	disease:疟	1112	1115	181	1	0.0141	0.0081	0.0713	71	
1368	disease:疮	3570	2838	429	1	0.012	0.0033	0.0111	83	
1368	disease:痈	1706	1425	274	1	0.004	0.0039	0.0143	253	
1368	disease:痹	2195	1304	300	1	0.0061	0.0034	0.0603	165	
1368	disease:霍乱	567	432	160	1	0.0909	0.0063	0.0411	11	formula:四逆汤 (rank 11, 2 later books)
1368	symptom:下利	1609	1299	271	1	0.0083	0.0052	0.0469	120	
1368	symptom:发热	2486	2910	267	4	0.0211	0.0213	0.0389	53	
1368	symptom:呕吐	1452	1207	341	4	0.012	0.0085	0.02	80	
1368	symptom:咳	2359	1902	339	1	0.0179	0.0063	0.016	56	
1368	symptom:喘	2198	1665	309	3	0.6818	0.678	0.024	1	herb:甘草 (rank 1, 2 later books); herb:人参 (rank 2, 2 later books)
1368	symptom:头痛	1550	1280	215	3	0.0471	0.0138	0.0417	9	formula:大承气汤 (rank 9, 2 later books)
1368	symptom:寒热	2427	1829	317	3	0.0176	0.0089	0.018	63	
1368	symptom:小便不利	779	466	147	1	0.0222	0.0078	0.0396	45	
1368	symptom:心烦	933	506	201	1	0.0357	0.0063	0.0359	28	
1368	symptom:恶寒	1296	1279	173	1	0.0098	0.006	0.0125	102	
1368	symptom:拘急	511	309	216	1	0.0047	0.0076	0.041	213	
1368	symptom:汗出	2980	1815	282	1	0.011	0.0044	0.0174	91	
1368	symptom:浊	646	1070	144	2	0.0215	0.0153	0.0351	52	
1368	symptom:淋	943	934	232	1	0.0078	0.0049	0.0119	129	
1368	symptom:渴	2664	2560	330	5	0.0156	0.0115	0.0224	124	
1368	symptom:腹满	1368	597	267	2	0.0387	0.0074	0.0209	15	formula:桂枝汤 (rank 15, 2 later books)
1368	symptom:腹痛	1487	1329	291	3	0.5159	0.5065	0.0234	1	herb:当归 (rank 1, 2 later

Cutoff	Topic	Past passages	Future passages	Candidates	Came true	AP Adamic–Adar	AP popularity	AP chance	First true link (Adamic–Adar rank)	True links in Adamic–Adar's top 20
										books); herb:人参 (rank 6, 2 later books); formula:四逆汤 (rank 14, 2 later books)
1368	symptom: 谵语	449	409	65	1	0.02	0.0213	0.0584	50	
1368	symptom: 身疼痛	380	365	79	1	0.25	0.0435	0.1242	4	formula:白虎汤 (rank 4, 2 later books)

Supplementary Figure



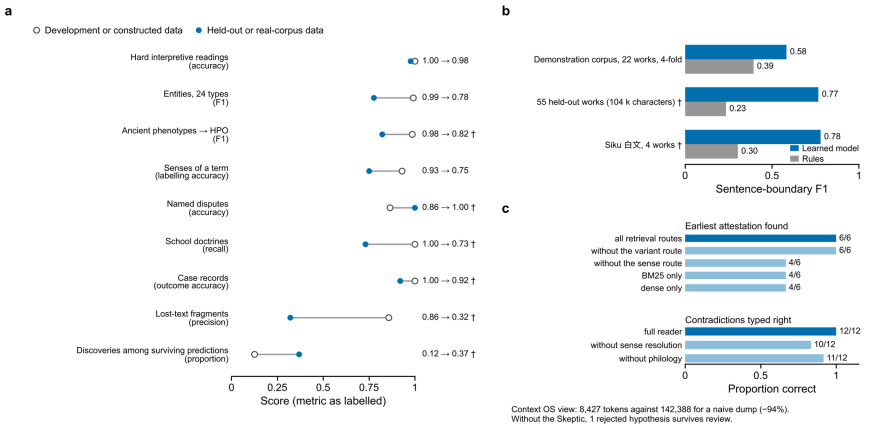
c

Failure	Where	What it did	What would catch it
Source families over-merged	H5, H7, H12, H13 (Fig. 3d)	the transitive closure chains 233 of 553 books into one family; with transcription links only, the four would rest on two or three families, but H5 and H13 rest on copies of one Jingui clause each, so one family is right for them	judge independence per claim, from the passages that support it, not per book
Co-listing read as association	R1; H8	R1: 口熬 and 上氣 are two items of the kidney channel's list of illnesses, which later books copy; H8: 小腹痛 belongs to the 尺脈伏 sentence, which gives 大平胃丸, not to 肾气丸	require the relation to be stated (indication, cause), not co-mention, before a lost or restated link counts
Concept drift on loose evidence	H2–H4	the evidence includes passages that do not name 中风 (迎風, 風氣百疾) and one Shanghan lun clause (傷寒中風) in four witnesses; the drifts' own permutation tests fail ($P = 0.66, 0.89, 0.98$), yet narrowing to 'exploratory' closes the objection and all three reach the expert	take drift evidence only from passages naming the term, copies once; make a failed test of the hypothesis's own statistic a hard objection
Case records read as doctrine	H9–H11	the 内科摘要 sentence and the 外科精要 passage are Xue Ji's records of treatments that failed (不應; 愈甚), not views; the 外科精要 and 校注妇人良方 witnesses are Xue Ji's editions (1547), whose notes and cases take the dates of the base texts (1263, 1248), so H9 sets Xue Ji against himself	read case records as records of practice; date editors' notes and cases as a later layer
Overlapping hypotheses	H2–H4; H9–H11	three drifts share the four witnesses of one clause, and three contradictions share one 内科摘要 passage	group hypotheses whose evidence overlaps before they are reviewed
Coverage of the open corpus	sources dossier	the open corpus keeps 8 of 24 chapters of 小儿药证直诀 (1119), none with 地黄丸, so the earliest composition of 六味地黄丸 found is of 1564; the 地黄丸 of 978–992 is another formula of the same name	state the sampled share beside every earliest attestation; rerun on the full store

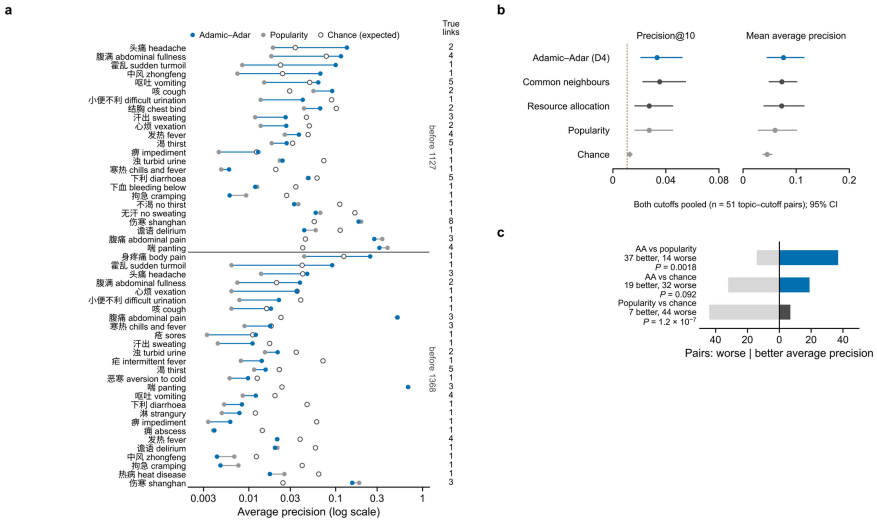
Also reported, not repaired: the two unsafe readings of the held-out interpretive cases (Fig. 4c) and the open-corpus benchmark, too small to separate the ranking from chance (its replication of Fig. 5).

Supplementary Fig. 1 | Where the end-to-end run of Fig. 3 went wrong, and why. a, Books merged into one source family under three rules: the harness's (the same work or author, or any transcription or rephrasing link between their passages, closed transitively), transcription links only, and the same work or author only. Under the harness's rule 36 of the 46 books that supplied evidence, and 233 of the 553 books of the open corpus, fall into one family. **b**, The independent families each final hypothesis rests on under the three rules; dashed line, the two that G5 requires. The hypotheses that G5 failed and that rest on two to three families once rephrasing links are left out (H5, H7, H12, H13) include two (H5, H13) whose evidence is copies of one clause, for which one family is right. **c**, The failure modes found by reading every surviving hypothesis against its evidence, with what each did and the check that would catch it. None is repaired here: the run is reported as it happened.

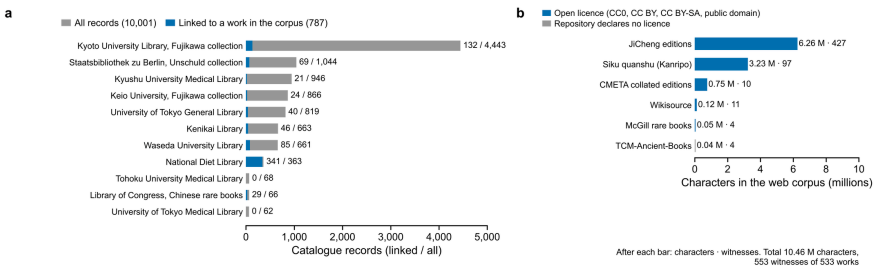
Supplementary Figures 2–4 and 5



Supplementary Fig. 2 | Where the rules transfer and where they do not. a, Development or constructed data (open circles) against held-out or real-corpus data (blue) for nine capabilities; numbers, the two scores; †, real-corpus value from the full corpus (`docs/evals.md`). Development sets were written with the rules they test. **b**, Sentence-boundary F1 of the learned punctuation model and of the rule baseline on the demonstration corpus (four folds), on 55 works held out of training, and on four unpunctuated *Siku* texts scored against their JiCheng punctuation (†, full corpus); every character was preserved in all settings. **c**, Ablations on the demonstration corpus: earliest-attestation accuracy with retrieval routes removed (6 terms), contradiction typing without sense resolution or philology (12 passage pairs), the context supplied to agents, and review without the Skeptic.

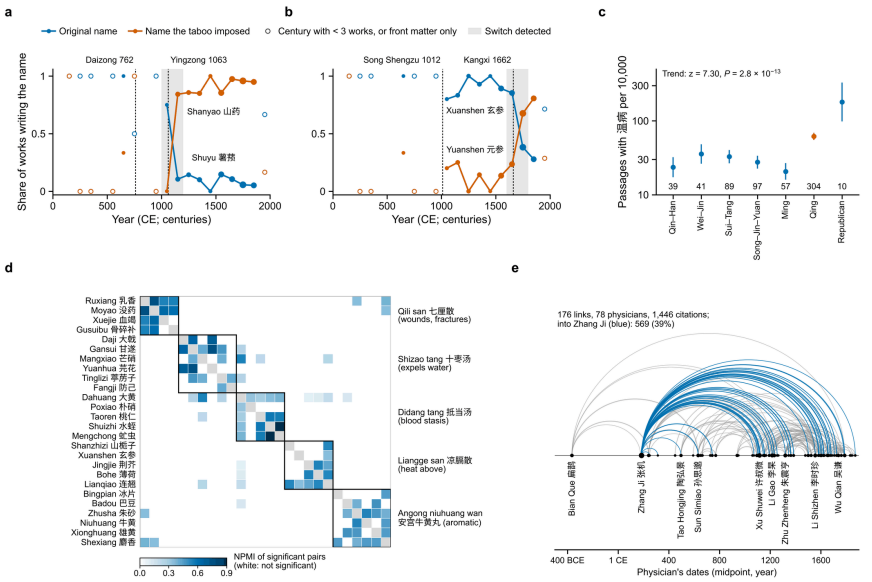


Supplementary Fig. 3 | The research-value benchmark on the open canon corpus. The benchmark of Fig. 5 re-run with identical rules on the open web corpus (533 works, most sampled by chapters). **a**, Average precision of Adamic–Adar (blue), popularity (grey) and the expectation under chance (open circles) in each of the 51 topic–cutoff pairs in which at least one link came true (24 at 1127, 27 at 1368; log scale); lines join Adamic–Adar to popularity, blue where Adamic–Adar is higher; right, links that came true in the pair. More than half of the pairs have a single true link (28 of 51; 109 among 10,115 candidates), so one hit moves a ranking. **b**, Precision@10 and mean average precision of the five rankers, both cutoffs pooled (n = 51 pairs); dots, means; lines, 95% confidence intervals; dashed line, base rate (1.08%). **c**, Paired comparisons of average precision, two-sided sign tests. Adamic–Adar still ranks above popularity (37 of 51 pairs, P = 0.0018) but cannot be told from chance (19 better, 32 worse, P = 0.092), and popularity falls below chance (7 better, 44 worse, P = 1.2×10^{-7}). The benchmark needs the full corpus; the replication is reported as a limit of the open subset. Every pair and every prediction: Supplementary Tables 11 and 12.



Supplementary Fig. 4 | The witnesses behind the texts. a, Catalogue records of printed and manuscript copies in eleven digitized collections (title, date, print or manuscript, shelfmark, IIIF manifest and licence; no image is downloaded) and those linked by title to a work of the corpus (787 records of 179 works); links made by title need checking by hand. **b**, Sources of

the open web corpus in millions of characters and witnesses. Blue, openly licensed transcriptions: JiCheng, whose selection, arrangement, punctuation and notes on public-domain texts are released under CC0 1.0 and its own writing under CC BY 4.0 (the modern editors' part of the few texts it took from modern reconstructions of lost books excepted); CMETA, CC BY 4.0; Kanseki Repository and Wikisource, CC BY-SA 4.0; McGill, public domain. Grey, the four TCM-Ancient-Books texts that match no JiCheng text, whose repository declares no licence, carried on the decision of the releasing institute (IMPF-AI). Sources whose terms forbid redistribution are never included. IIIF, International Image Interoperability Framework.



Supplementary Fig. 5 | On the open canon corpus, the study layer recovers history written into the texts. All panels are computed by the deterministic study layer on the open web corpus (533 works, 168,278 passages). **a**, **b**, Share of works per century writing an original drug name and the name a naming taboo imposed: 薯蓣 and 山药 (**a**), 玄参 and 元参 (**b**). Dashed lines, the years from which the names were avoided: Tang Daizong (李豫, 762) and Song Yingzong (赵曙, 1063) in **a**; the Song ancestor Shengzu (赵玄朗, 1012) and the Kangxi emperor (玄烨, 1662) in **b**; shading, the switch interval the tool detected; marker area, works in the century; open markers, centuries with fewer than three works or only undated front matter dated conservatively to 1911. A work writing both names counts for both. **c**, Passages mentioning 温病 (warm disease) per 10,000 passages by period, with Wilson 95% confidence intervals; numbers, passages with the term. Two-sided Cochran–Armitage test for trend across the ordered periods. **d**, Normalized pointwise mutual information (NPMI) of drug pairs co-occurring significantly in 4,554 compositions (a composition repeated in copies of one work counted once; pairs in at least ten works tested; one-sided Fisher exact test, Benjamini–Hochberg q < 0.05; 1,184 pairs). Groups are communities of the strong pairs (NPMI ≥ 0.3), found without formula names; shown (outlined) are the five of twelve whose drugs hold together most tightly (mean NPMI of their significant pairs). Each holds the core of a classical formula (right); liquorice (甘草), present in 43% of compositions, joins none. **e**, Citations between physicians: arcs join citing and cited physicians placed at the midpoints of their dates; width ∝ √citations; blue, citations of Zhang Ji (张机). Five links whose cited physician lived after the citing one (28 citations) are omitted as chronologically impossible.

Supplementary Table 13 and 14

Supplementary Table 13 | Sources of the corpus

Source	Licence	Works admitted	Characters (millions)	Transcription and notes
Kanseki Repository KR3e, Siku quanshu medical section (漢籍リポジトリ)	CC BY-SA 4.0	100	25.66*	Wenyuange Siku and Sibu congkan facsimiles; unpunctuated; catalogue with authors, dates, layers
JiCheng 笈成 edited texts, v1.4.8	CC0 1.0 (editorial work); base texts public domain§	816	93.64*	Volunteer transcriptions with modern punctuation; 41 of 857 works excluded
CMETA 數位中醫校書郎 collated editions	CC BY 4.0 (collation); base texts public domain	11	1.49	Collated page by page against images; passages carry page numbers and image links
Tōa Igaku Kyōkai 東亜医学協会 texts	© the society; local research only†	7	0.33	Suwen, Lingshu, Nanjing, Shanghan lun, Jingui yaolüe, Shennong bencao jing, Bian Que biography
Wikisource (Category 中醫, 醫書)	CC BY-SA 4.0	18	0.77	Works new after de-duplication of 627 candidates
McGill University Library rare books	Public domain	5	0.40	Qing prints and manuscripts on gynaecology
TCM-Ancient-Books; tcmoc	Not declared by the repositories‖	8	0.37	Mostly simplified copies of JiCheng texts, de-duplicated work by work
Hugging Face classical-tcm-canon	Proprietary-commercial; local research only†	1	0.31	One work (Zhang Zhicong's commentary on the Shanghan lun)
Kanseki Repository catalogue (KR-Catalog)	CC BY-SA 4.0	–	–	Roles and life dates of 132 persons responsible for the Siku works
Image catalogues of 11 collections‡	Recorded per collection	–	–	10,001 records (787 linked to 179 works): title, date, print or manuscript, shelfmark, IIIF manifest
Total		966	≈123	2.05 M passages; 761 distinct works; analysis periods 475 BCE–1949

*Counted in the corpus store (README); the catalogues of these two sources do not record characters. Other counts sum the catalogue records of the admitted works.

†Used for local research only and never redistributed; the open web corpus carries none of the texts whose terms forbid redistribution.

§JiCheng releases its selection, arrangement, punctuation and notes on public-domain texts under CC0 1.0 and its own writing under CC BY 4.0 (https://jicheng.tw/tcm/copyright.html, consulted 29 September 2026); the open web corpus carries 427 JiCheng witnesses under these terms, crediting JiCheng. The modern editors' part of the few texts JiCheng transcribed from modern reconstructions of lost books is not covered.

‖Mostly simplified-character copies of JiCheng texts; de-duplication removes every copy, and the 8 texts kept match no JiCheng text. The 4 in the open web corpus are carried on the decision of the releasing institute (IMPF-AI); the others are used for local research only.

‡Kyoto University (Fujikawa), Staatsbibliothek zu Berlin (Unschuld), Kyushu University, Keio University (Fujikawa), University of Tokyo (two libraries), Kenikai, Waseda University, National Diet Library, Tohoku University and Library of Congress. Only metadata are held; no image is downloaded.

Sources are ranked by trust for de-duplication (Kanripo, JiCheng, McGill, CMETA, AEAM, Wikisource, web collections); a candidate sharing enough normalized 12-character shingles with an admitted work is recorded as its copy, while McGill, CMETA and AEAM transcriptions are kept as independent witnesses. IIIF, International Image Interoperability Framework.

Supplementary Table 14 | Evaluation of TaoChronos across its three layers

Layer	Suite	Data	Metric	Development or constructed	Held-out or real corpus
Boundaries	Invariants	19 probes of the harness under test	Held	19/19	Each of 6 boundaries removed breaks its invariant
Readings	Hard interpretive cases	107 cases in 8 classes	Accuracy (unsafe errors)	23/23 (0)	82/84 = 0.976 (2)
Research value†	Link anticipation	189 topic–cutoff pairs; 29,467 candidates	Precision@10	–	0.168 (popularity 0.134; chance 0.054); higher average precision than popularity in 149 of 189, P = 2.4 × 10^{-16}
	End-to-end predictions	Held-out topics, cutoff 1368	Discoveries among survivors	1/8	7/19 = 0.37; leakage 0
Verification	Fabricated quotations	60 quotations × 4 probes	Handled correctly	–	60/60 in every probe; variant spellings accepted
	Provenance	Committed claims and evidence	Verbatim	–	100%; mean chain depth 7.8 levels
	Claim extraction	42 gold hyperedges	Precision / recall	1.00 / 0.976	–
	Temporal and anachronism guards	Constraints, earliest attestations, equations	Correct	6/6 · 6/6 · 10/10	Leakage 0; agent-asserted equivalence blocked
Deep discovery	Collation	5 traditions copied down a known stemma	Variant-unit recall / precision	0.998 / 0.999	–
	Semantic reuse	33 typed pairs	Accuracy	0.97	–
	Stratigraphy	5 composites; the Suwen	Layer accuracy	0.955	Yunqi chapters 7/7 in the minor layer†
	Lost-text fragments	Constructed lost work; Qianjin yaofang	Precision	0.86	0.32†
Task specification	Punctuation	55 held-out works (104 k characters)†	Boundary F1	–	0.767 (rules 0.235); characters kept 1.000
	Drug pairs	66,196 compositions†	Textbook pairs significant	–	7/7, lift > 1.5
V2	Entities (24 types)	41 development, 25 held-out sentences	F1	0.991	0.776
	Ancient phenotypes → HPO	20 development cases; 12 random passages†	F1	0.985	0.822
	Evidence audit	19 claims	Verdicts right (false SUPPORTED)	19/19 (0)	–

Development and constructed gold sets were written by the authors with the rules they test; they are regression tests of intended behaviour, not unbiased benchmarks. Only held-out and real-corpus columns estimate generalization.

†Needs the full corpus store, which is not distributed; values from docs/evals.md. All other values were reproduced for this paper by `taochronos eval` on the demonstration corpus (40 suites, 47 s).

P value from a two-sided sign test of average precision over topic–cutoff pairs. HPO, Human Phenotype Ontology.

Supplementary Data

Supplementary Data 1 | The Discovery Report of the end-to-end run (`Supplementary_Data_1_e2e_discovery_report.md`): the report the harness wrote, in Chinese with English section titles, as the conversation client and the command line show it.

Supplementary Data 2 | The event log of the end-to-end run (`Supplementary_Data_2_e2e_event_log.jsonl.gz`): every event in order, one JSON object a line, each with its sequence number, type, actor, principal and payload; replaying it rebuilds the state from which Fig. 3 and the report were drawn.